Advances in Single-Qubit Phase Estimation

Paper presentation

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Outline

- Estimation tools
  - Case I
  - Case II
  - Case III

- Example applications for each case

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In the Shadow of the Hadamard Test: Using the Garbage State for Good and Further Modifications

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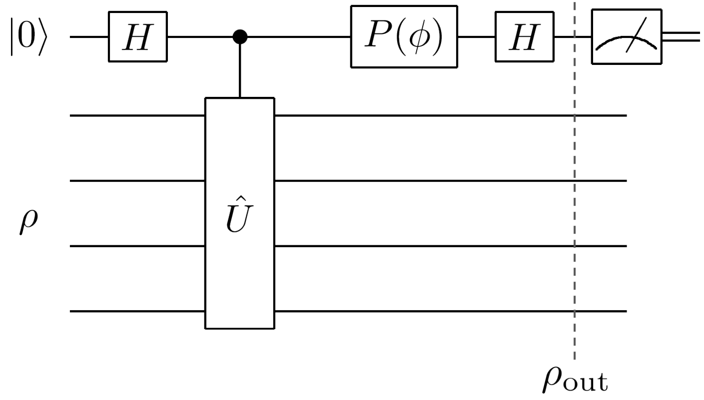
Basic Hadamard Test & Classical Shadows

$$p(0) = \frac{1}{2}\mathrm{tr}(\rho) + \frac{1}{2}\mathcal{R}e\left[e^{-i\phi}\mathrm{tr}(U\rho)\right]$$

$$p(1) = \frac{1}{2}\mathrm{tr}(\rho) - \frac{1}{2}\mathcal{R}e\left[e^{-i\phi}\mathrm{tr}(U\rho)\right]$$



See Appendix A

$$P(\phi) = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{bmatrix}$$

For $\phi = 0$

$$p(0) - p(1) = \mathcal{R}e\left[\mathrm{tr}(U\rho)\right]$$

For $\phi = \frac{\pi}{2}$

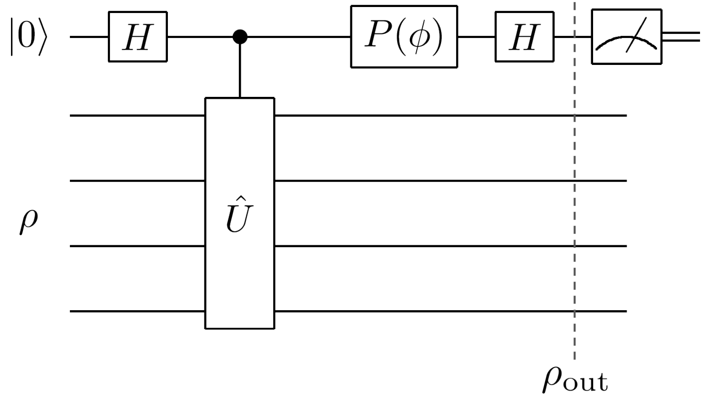
$$p(0) - p(1) = \mathcal{I}m\left[\mathrm{tr}(U\rho)\right]$$

Ancilla measurements capture the expectation value of the unitary.

The information in the *system's register* is discarded.

After measuring the auxiliary, system register is at a state...

$$\begin{aligned}\rho(Z) :&= \mathrm{tr}_{\mathrm{aux}}\left[(Z \otimes I_n)\rho_{\mathrm{out}}\right] \\ &= \frac{1}{2}\left(U\rho e^{i\phi} + \rho U^\dagger e^{-i\phi}\right)\end{aligned}$$



See Appendix B

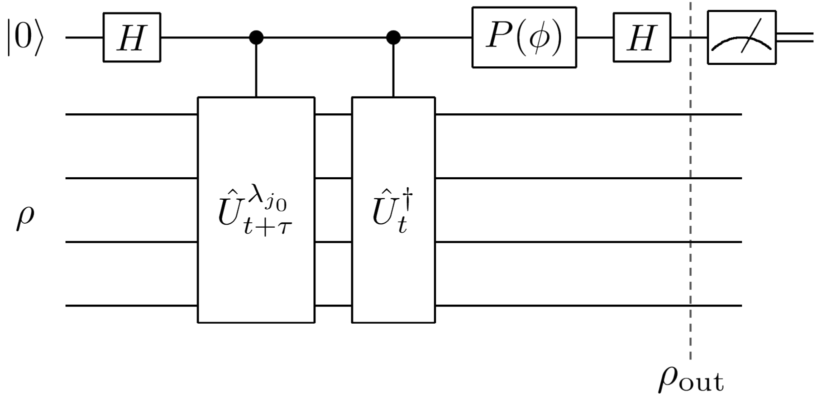
$$\boxed{\begin{aligned}\langle Z \otimes O\rangle_{\rho_{\mathrm{out}}} &= \mathrm{tr}[O\rho(Z)] \\ &= \mathcal{R}e\left[\mathrm{tr}(e^{i\phi}OU\rho)\right]\end{aligned}}$$

with **O** being hermitian.

We utilize the system register to obtain additional information in parallel with the Hadamard test.

Applying the approach on the Burger's equation case...

$$\rho(Z) = \frac{1}{2}\left(e^{-i\phi}\rho\,\hat{U}_{t+\tau}^{\lambda_{j0}\,\dagger}\,\hat{U}_t + e^{i\phi}\hat{U}_t^{\dagger}\,\hat{U}_{t+\tau}^{\lambda_{j0}}\,\rho\right)$$



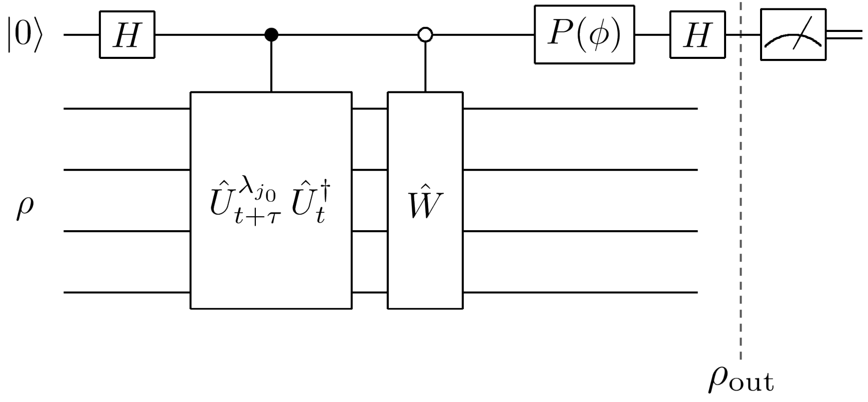
$$\begin{aligned}\langle Z \otimes O\rangle_{\rho_{\text{out}}} &= \mathrm{tr}\left[\hat{O}\,\rho(Z)\right] \\ &= \mathcal{R}e\left[\mathrm{tr}\left(e^{i\phi}\hat{O}\,\hat{U}_t^{\dagger}\,\hat{U}_{t+\tau}^{\lambda_{j0}}\,\rho\right)\right]\end{aligned}$$

Where $\hat{U}_{t+\tau}^{\lambda_{j0}}$ is the ansatz at time-step $t+\tau$ with fixed parameter $\lambda_j = \lambda_{j0}$

Real and imaginary parts are estimated (with different φ) leaving the ancilla intact.

Adding an anti-controlled unitary **W**

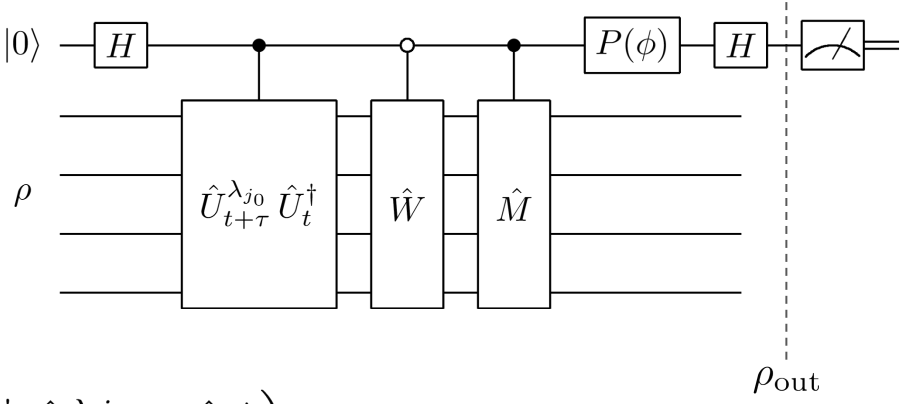
$$\rho(Z) = \frac{1}{2}\left(\hat{U}_{t+\tau}^{\lambda_{j_0}}\,\hat{U}_t^\dagger\,\rho\,\hat{W}^\dagger + \hat{W}\,\rho\,\hat{U}_t\,\hat{U}_{t+\tau}^{\lambda_{j_0}\,\dagger}\right)$$



By adding **W** we can further modify the state **ρ** upon we measure.

$$\langle Z \otimes O\rangle_{\rho_{\text{out}}} = \mathcal{R}e\left[\,\mathrm{tr}\left(\hat{W}^\dagger\,\hat{O}\,\hat{U}_{t+\tau}^{\lambda_{j_0}}\,\hat{U}_t^\dagger\,\rho\right)\right]$$

Adding a second controlled unitary **M** together with **W**



$$\rho(Z) = \frac{1}{2}\left(e^{-i\phi}\hat{W}\,\rho\,\hat{U}^{\lambda_{j_0}\,\dagger}_{t+\tau}\,\hat{U}_t\,\hat{M}^{\dagger} + e^{i\phi}\hat{M}\,\hat{U}^{\dagger}_{t}\,\hat{U}^{\lambda_{j_0}}_{t+\tau}\,\rho\,\hat{W}^{\dagger}\right)$$

By adding **W** and **M** we gain complete freedom on the state **ρ** upon we measure.

$$\langle Z \otimes O\rangle_{\rho_{\text{out}}} = \mathcal{R}e\left[\operatorname{tr}\left(e^{-i\phi}\,\hat{U}^{\lambda_{j_0}\,\dagger}_{t+\tau}\,\hat{U}_t\,\hat{M}^{\dagger}\,\hat{O}\,\hat{W}\,\rho\right)\right]$$

Estimation Tools

Rearranging by utilizing the cyclic property of the trace. With $\hat{O}$ hermitian and $\hat{M}, \hat{W} \in U(2^n)$.

$$\langle Z \otimes O \rangle_{\rho_{\mathrm{out}}} = \mathcal{R}e\left[\mathrm{tr}\left(e^{i\phi}\hat{O}\,\hat{U}_t^{\dagger}\,\hat{U}_{t+\tau}^{\lambda_{j_0}}\,\rho\right)\right]$$

$$\langle Z \otimes O \rangle_{\rho_{\mathrm{out}}} = \mathcal{R}e\left[\mathrm{tr}\left(\hat{O}\,\hat{U}_{t+\tau}^{\lambda_{j_0}}\,\hat{U}_t^{\dagger}\,\rho\,\hat{W}^{\dagger}\right)\right]$$

$$\langle Z \otimes O \rangle_{\rho_{\mathrm{out}}} = \mathcal{R}e\left[\mathrm{tr}\left(e^{-i\phi}\hat{O}\,\hat{W}\,\rho\,\hat{U}_{t+\tau}^{\lambda_{j_0}\,\dagger}\,\hat{U}_t\,\hat{M}^{\dagger}\right)\right]$$

Goal: measure observables with physical interest on states prepared by useful unitaries which don't affect the optimization process.

The different cases provide increasing freedom on the state upon which we measure $\hat{O}$, with the tradeoff of additional circuit depth.

Potential Uses (case I)

For $\hat{O} = I_n$ and $\rho = |\mathbf{0}\rangle\langle\mathbf{0}|_n$

$$\langle Z \otimes O \rangle = \mathrm{Re}\left[\mathrm{tr}\left(e^{i\phi}\left|\Psi_{t+\tau}^{\lambda_{j_0}}\right\rangle\left\langle\Psi_t\right|\right)\right]$$

An estimator for the fidelity. Essentially measuring the change between the previous and current variational states.

Potential Uses (case II)

Let $\hat{U}_{t+\tau}^{\lambda_{j_0}} \hat{U}_t^\dagger = e^{i\hat{H}t_1}$ and $\hat{W} = e^{i\hat{H}t_2}$ for some appropriate Hamiltonian $\hat{H}$.

$$\mathrm{tr}\left[\rho(Z)\right] = \alpha^2 \cos\left[E_1(t_1 - t_2)\right] + \beta^2 \sin\left[E_2(t_1 - t_2)\right]$$

$$\hat{H}\left|\mu\right\rangle = E_1\left|\mu\right\rangle$$

$$\hat{H}\left|\nu\right\rangle = E_2\left|\nu\right\rangle$$

$$\left|\psi\right\rangle = \alpha\left|\mu\right\rangle + \beta\left|\nu\right\rangle$$

The system register is prepared in some state $\rho = \left|\psi\right\rangle\left\langle\psi\right|$.

Potential Uses (case III)

For $\rho = |\mathbf{0}\rangle\langle\mathbf{0}|_n$, $\hat{M} = \hat{U}_t$ and $\hat{W} = \hat{U}_{t+\tau}^{\lambda_{j0}}$.

$$\langle Z \otimes O\rangle_{\rho_{\text{out}}} = \mathcal{R}e \ \mathrm{Tr}\left(e^{-i\phi}\hat{O}\,\hat{\rho}_{t+\tau}^{\lambda_{j0}}\right)$$

For $\rho = |\mathbf{0}\rangle\langle\mathbf{0}|_n$, $\hat{M} = \hat{U}_t, \hat{W}|\mathbf{0}\rangle = |\Phi\rangle$ and $\hat{O} = |\Phi\rangle\langle\Phi|$.

$$\langle Z \otimes O\rangle_{\rho_{\text{out}}} = \mathcal{R}e\left(e^{-i\phi}\left\langle \Psi_{t+\tau}^{\lambda_{j0}} \middle| \Phi\right\rangle\right)$$

with $|\Phi\rangle$ some (preparable) pure state.

Things to look into…

- Utilization along with gradient-based optimizers.
- Investigate potential uses of the state $\hat{U}_t^\dagger \left|\mathbf{0}\right\rangle$ for case II.
- In slide 10, consider non-eigenstate |ψ⟩ but with eigenstate overlap (working on that).
- Integrate QEM techniques between unitaries (working on that).